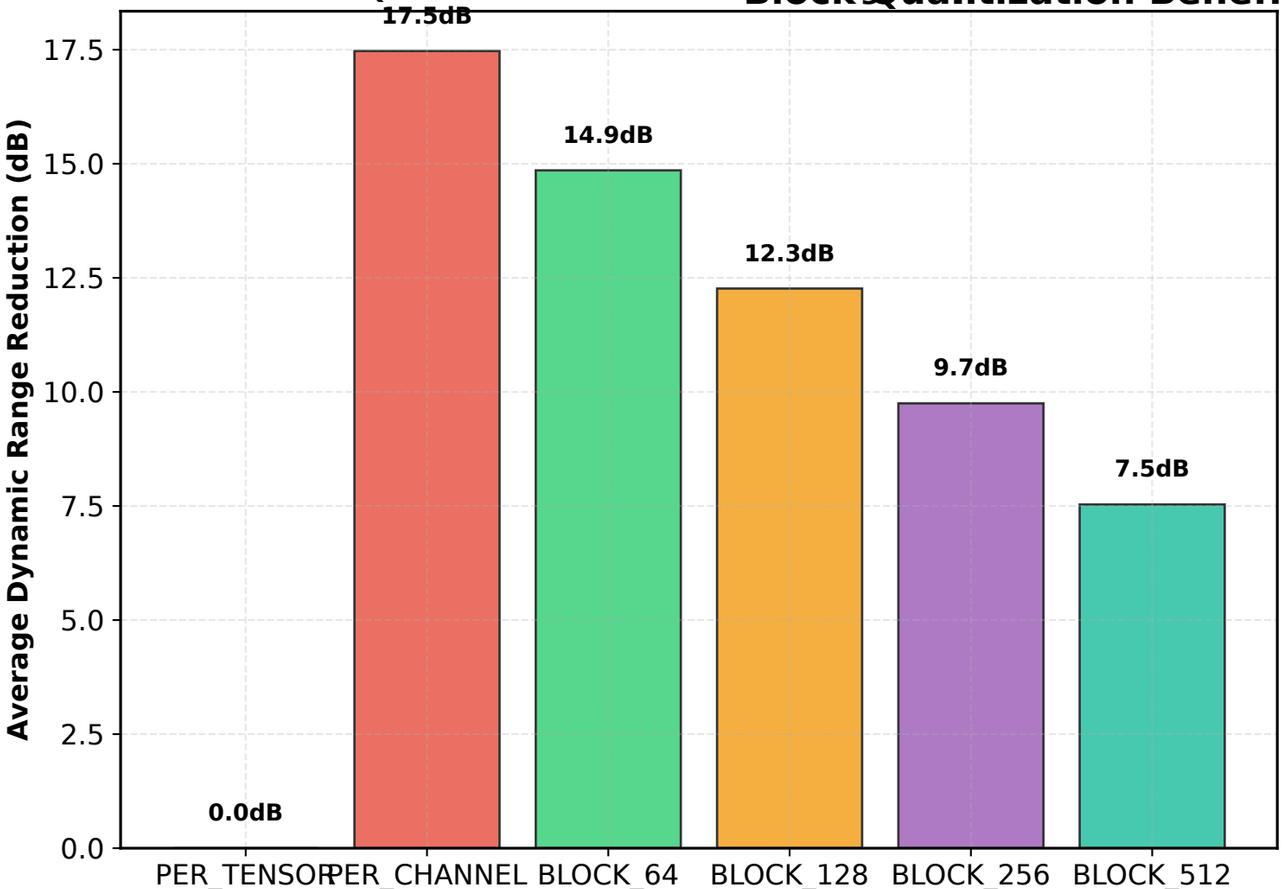
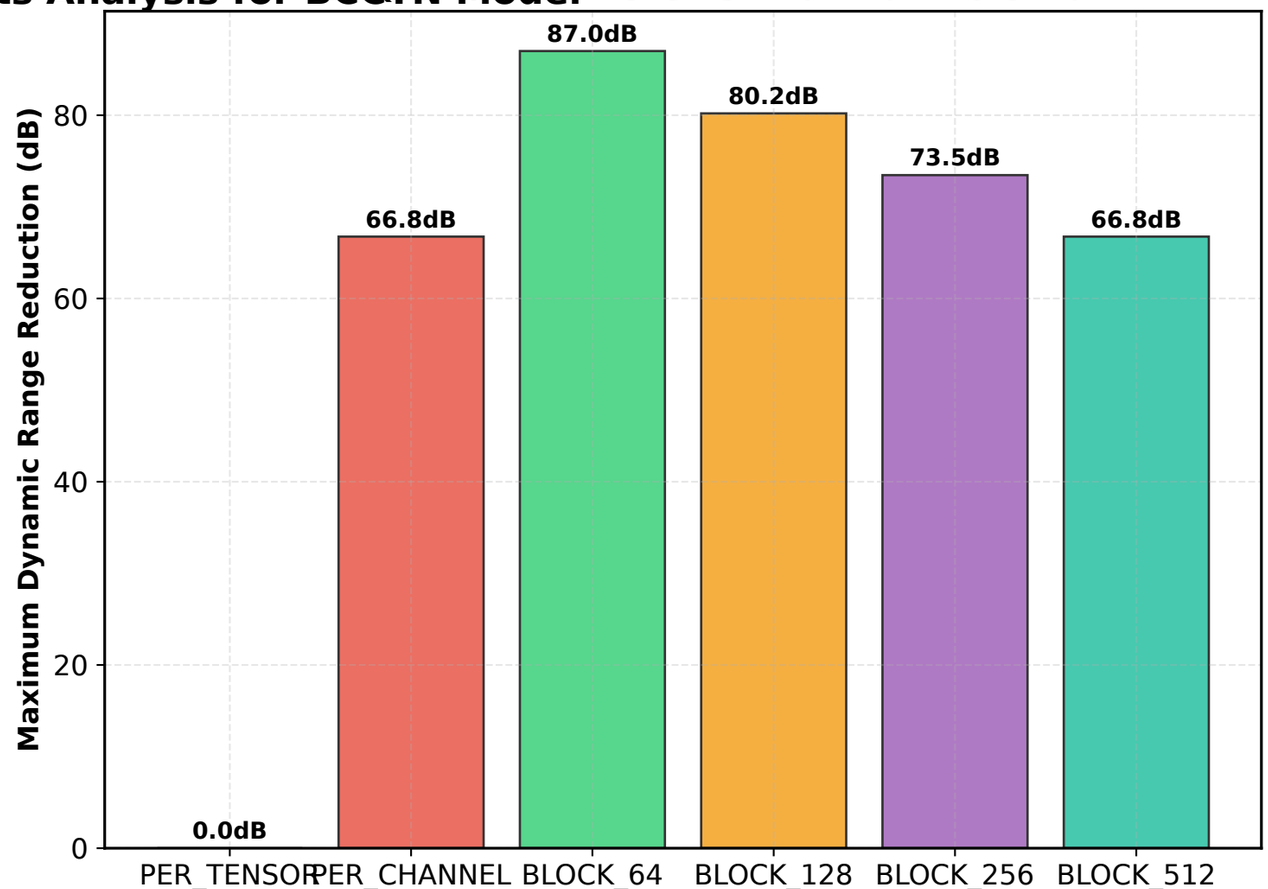
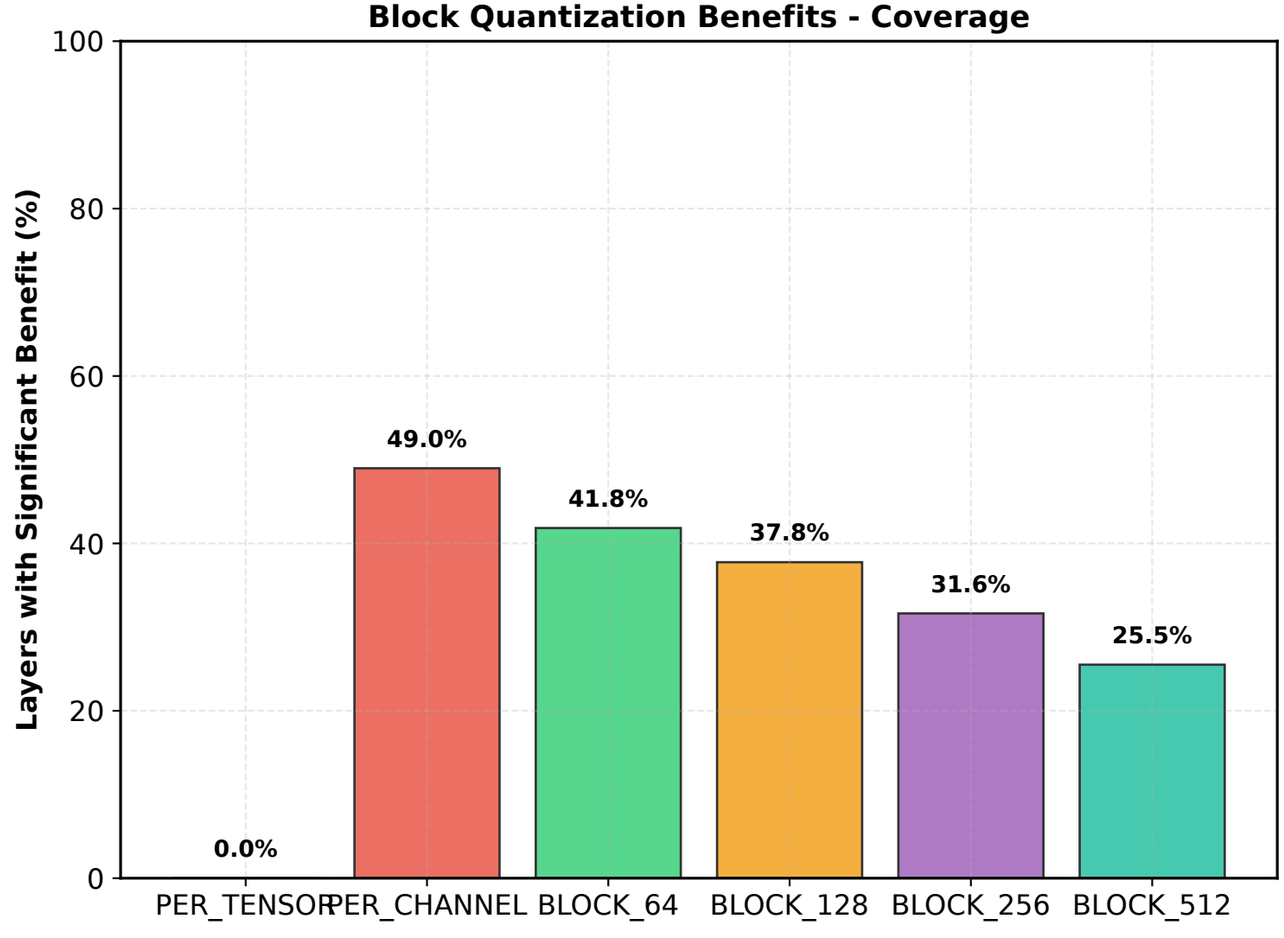
Block Quantization Benefits Analysis for BCCTN Model







BLOCK QUANTIZATION RECOMMENDATIONS

BEST OVERALL: PER_CHANNEL
- **Average improvement: 17.5 dB**
- **Coverage: 49.0% of layers benefit**
- **Description: Per output channel**

CONFIGURATION GUIDE:
- **PER_TENSOR: Simplest, but limited for high dynamic range**
- **PER_CHANNEL: Good balance for conv layers**
- **BLOCK_64/128: Best for very high dynamic range layers**
- **BLOCK_256/512: Moderate improvement, easier implementation**

IMPLEMENTATION PRIORITY:
1. **Try PER_CHANNEL first**
2. **Fall back to PER_CHANNEL for compatibility**
3. **Use PER_TENSOR only if others fail**